

```
CREATE DATABASE sql_assignment1;
```

```
USE sql_assignment1;
```

```
CREATE TABLE Employees (  
  EmpID INT PRIMARY KEY,  
  Name VARCHAR(50) NOT NULL,  
  Dept VARCHAR(20) NOT NULL,  
  Role VARCHAR(20) NOT NULL,  
  Salary INT NOT NULL,  
  Gender CHAR(1) NOT NULL,  
  JoinDate DATE NOT NULL,  
  Address VARCHAR(100) NOT NULL,  
  City VARCHAR(50) NOT NULL,  
  State CHAR(2) NOT NULL,  
  Region VARCHAR(10) NOT NULL  
);
```

```
INSERT INTO Employees (EmpID, Name, Dept, Role, Salary, Gender, JoinDate, Address, City, State,  
Region) VALUES
```

```
(1, 'Alice', 'HR', 'Manager', 55000, 'F', '2019-03-15', '101 Maple St', 'New York', 'NY',  
'East'),
```

```
(2, 'Bob', 'IT', 'Developer', 72000, 'M', '2020-07-22', '22 Oak Ave', 'Chicago', 'IL',  
'Midwest'),
```

```
(3, 'Carol', 'Finance', 'Analyst', 68000, 'F', '2018-11-01', '303 Pine Rd', 'New York', 'NY',  
'East'),
```

```
(4, 'David', 'HR', 'Executive', 52000, 'M', '2021-01-10', '44 Birch Ln', 'Chicago', 'IL',  
'Midwest'),
```

```
(5, 'Eve', 'IT', 'Senior Dev', 85000, 'F', '2017-06-30', '55 Cedar Blvd', 'Houston', 'TX',  
'South'),
```

```
(6, 'Frank', 'Finance', 'Manager', 74000, 'M', '2019-09-14', '606 Elm St', 'New York', 'NY',
```

'East'),
(7, 'Grace', 'Marketing', 'Executive', 60000, 'F', '2022-02-28', '77 Spruce Dr', 'Chicago', 'IL',
'Midwest'),
(8, 'Hank', 'IT', 'Lead', 90000, 'M', '2016-04-05', '88 Walnut Way', 'Houston', 'TX',
'South'),
(9, 'Ivy', 'HR', 'Executive', 58000, 'F', '2020-12-19', '99 Ash Ct', 'New York', 'NY',
'East'),
(10, 'Jack', 'Finance', 'Analyst', 65000, 'M', '2021-03-07', '10 Willow Pl', 'Chicago', 'IL',
'Midwest'),
(11, 'Karen', 'Marketing', 'Manager', 62000, 'F', '2018-08-23', '11 Poplar St', 'Houston', 'TX',
'South'),
(12, 'Leo', 'IT', 'Developer', 78000, 'M', '2019-11-11', '12 Sycamore Ave', 'New York', 'NY',
'East'),
(13, 'Mia', 'HR', 'Executive', 53000, 'F', '2022-05-16', '13 Hickory Rd', 'Chicago', 'IL',
'Midwest'),
(14, 'Nate', 'Finance', 'Manager', 71000, 'M', '2017-12-02', '14 Chestnut Ln', 'Houston',
'TX', 'South'),
(15, 'Olivia', 'Marketing', 'Analyst', 66000, 'F', '2020-06-18', '15 Magnolia Dr', 'New York',
'NY', 'East'),
(16, 'Paul', 'IT', 'Senior Dev', 82000, 'M', '2018-03-27', '16 Dogwood Blvd', 'Chicago', 'IL',
'Midwest'),
(17, 'Quinn', 'HR', 'Executive', 57000, 'F', '2021-09-09', '17 Hawthorn St', 'Houston', 'TX',
'South'),
(18, 'Rita', 'Finance', 'Lead', 76000, 'F', '2019-01-25', '18 Redwood Ave', 'New York', 'NY',
'East'),
(19, 'Sam', 'Marketing', 'Executive', 63000, 'M', '2022-07-14', '19 Cypress Ct', 'Chicago', 'IL',
'Midwest'),
(20, 'Tina', 'IT', 'Lead', 88000, 'F', '2016-10-31', '20 Juniper Way', 'Houston', 'TX',
'South'),
(21, 'Uma', 'HR', 'Manager', 60000, 'F', '2020-04-03', '21 Aspen Pl', 'Dallas', 'TX',
'South'),

(22, 'Victor', 'Finance', 'Analyst', 69000, 'M', '2018-07-19', '22 Cottonwood St', 'Dallas', 'TX', 'South'),

(23, 'Wendy', 'Marketing', 'Manager', 64000, 'F', '2021-11-27', '23 Palmetto Ave', 'Phoenix', 'AZ', 'West'),

(24, 'Xander', 'IT', 'Architect', 95000, 'M', '2015-02-14', '24 Mesquite Rd', 'Phoenix', 'AZ', 'West'),

(25, 'Yara', 'HR', 'Executive', 54000, 'F', '2022-09-08', '25 Saguaro Ln', 'Phoenix', 'AZ', 'West'),

(26, 'Zoe', 'Finance', 'Manager', 73000, 'F', '2019-05-21', '26 Palo Verde Dr', 'Dallas', 'TX', 'South'),

(27, 'Aaron', 'Marketing', 'Analyst', 67000, 'M', '2020-10-15', '27 Ocotillo Blvd', 'Phoenix', 'AZ', 'West'),

(28, 'Bella', 'IT', 'Senior Dev', 80000, 'F', '2017-08-06', '28 Ironwood St', 'Chicago', 'IL', 'Midwest'),

(29, 'Carlos', 'HR', 'Executive', 56000, 'M', '2021-06-24', '29 Pecan Ave', 'Houston', 'TX', 'South'),

(30, 'Diana', 'Finance', 'Lead', 77000, 'F', '2018-02-09', '30 Peachtree Ct', 'New York', 'NY', 'East'),

(31, 'Ethan', 'IT', 'Developer', 74000, 'M', '2019-04-18', '31 Magnolia St', 'Dallas', 'TX', 'South'),

(32, 'Fiona', 'Marketing', 'Executive', 61000, 'F', '2022-01-05', '32 Bluebonnet Ave', 'Dallas', 'TX', 'South'),

(33, 'George', 'Finance', 'Analyst', 66000, 'M', '2020-08-11', '33 Sunflower Rd', 'Phoenix', 'AZ', 'West'),

(34, 'Helen', 'HR', 'Manager', 59000, 'F', '2017-03-22', '34 Cactus Ln', 'Phoenix', 'AZ', 'West'),

(35, 'Ivan', 'IT', 'Lead', 92000, 'M', '2015-09-30', '35 Desert Dr', 'Phoenix', 'AZ', 'West'),

(36, 'Julia', 'Marketing', 'Manager', 65000, 'F', '2019-12-03', '36 Mesa Blvd', 'Dallas', 'TX', 'South'),

(37, 'Kevin', 'Finance', 'Manager', 70000, 'M', '2021-07-20', '37 Prairie St', 'Chicago', 'IL',

'Midwest'),
 (38, 'Laura', 'HR', 'Executive', 55000, 'F', '2022-10-14', '38 Lakeview Ave', 'Chicago', 'IL',
 'Midwest'),
 (39, 'Mark', 'IT', 'Developer', 76000, 'M', '2018-05-07', '39 Riverview Rd', 'New York', 'NY',
 'East'),
 (40, 'Nina', 'Marketing', 'Analyst', 63000, 'F', '2020-03-25', '40 Bayview Ln', 'New York',
 'NY', 'East'),
 (41, 'Oscar', 'Finance', 'Analyst', 67000, 'M', '2019-08-19', '41 Hillside Dr', 'Dallas', 'TX',
 'South'),
 (42, 'Priya', 'HR', 'Senior Dev', 79000, 'F', '2016-11-28', '42 Creekside Blvd', 'Houston', 'TX',
 'South'),
 (43, 'Raj', 'IT', 'Manager', 85000, 'M', '2018-06-14', '43 Brookside St', 'Phoenix', 'AZ',
 'West'),
 (44, 'Sarah', 'Marketing', 'Lead', 71000, 'F', '2017-02-01', '44 Springdale Ave', 'New York',
 'NY', 'East'),
 (45, 'Tom', 'Finance', 'Senior Dev', 81000, 'M', '2020-09-09', '45 Fairview Rd', 'Chicago', 'IL',
 'Midwest'),
 (46, 'Uma', 'IT', 'Architect', 97000, 'F', '2014-12-22', '46 Northgate Ln', 'New York', 'NY',
 'East'),
 (47, 'Vera', 'Marketing', 'Manager', 68000, 'F', '2021-04-30', '47 Eastwood Dr', 'Dallas', 'TX',
 'South'),
 (48, 'Will', 'HR', 'Analyst', 51000, 'M', '2022-11-15', '48 Westfield Blvd', 'Phoenix', 'AZ',
 'West'),
 (49, 'Xena', 'Finance', 'Lead', 78000, 'F', '2019-10-06', '49 Southpark St', 'Chicago', 'IL',
 'Midwest'),
 (50, 'Yusuf', 'IT', 'Senior Dev', 87000, 'M', '2016-07-17', '50 Central Ave', 'Houston', 'TX',
 'South');

select * from employees;

-- Q1. Count the total number of employees in each department.

select dept,count(*) as empcount from employees group by dept;

-- Q2. Find the total salary paid in each department.

select dept,sum(salary) as totalsalary from employees group by dept;

-- Q3. Find the average salary by gender.

select gender,avg(salary) as avgsalary from employees group by gender;

-- Q4. Find the maximum salary in each department.

select dept,max(salary) as maxsalary from employees group by dept;

-- Q5. Find the minimum salary in each role.

select role,min(salary) minsalary from employees group by role order by role;

-- Q6. Count the number of employees in each city

select city,count(*) as empcount from employees group by city;

-- Q7. Find the total salary paid per region.

select region,sum(salary) as totalsalary from employees group by region;

-- Q8. List departments sorted by their average salary (highest first).

select dept,avg(salary) as avgsalary from employees group by dept order by avgsalary desc;

-- Q9. List cities by total salary, ascending.

select city,sum(salary) as totalsalary from employees group by city order by totalsalary;

-- Q10. List roles sorted by employee count descending, then role name ascending.

select role,count(*) as empcount from employees group by role order by empcount desc,role;

-- Q11. List states with their employee count, sorted by count descending then state name.

select state,count(*) as empcount from employees group by state order by empcount desc,state desc;

-- Q12. Show the total salary per region, ordered from highest to lowest

select region, sum(salary) as totalsalary from employees group by region order by totalsalary desc;

-- Q13. List departments sorted by minimum salary descending.

select dept, min(salary) as minsalary from employees group by dept order by minsalary desc;

-- Q14. Show only departments with more than 10 employees.

select dept,count(*) as empcount from employees group by dept having empcount > 10 order by dept;

-- Q15. Show departments where the average salary exceeds 65000.

select dept,avg(salary) as avgsalary from employees group by dept having avgsalary > 65000;

-- Q16. Show cities where total salary exceeds 600000.

select city,sum(salary) as totalsalary from employees group by city having totalsalary > 600000;

-- Q17. Show roles with more than 7 employees.

select role,count(*) as empcount from employees group by role having empcount > 7;

-- Q18. Show regions where the average salary is below 70000.

select region,avg(salary) as avgsalary from employees group by region having avgsalary < 70000;

-- Q19. Show departments where the max salary is at least 90000.

select dept,max(salary) as maxsalary from employees group by dept having maxsalary > 90000;

-- Q20. Show states where total headcount is more than 12.

select state , count(*) as empcount from employees group by state having empcount > 12;

-- Q21. Show cities where the minimum salary is above 60000.

select city,min(salary) as minsalary from employees group by city having minsalary > 60000;

-- Q22. Count employees grouped by department AND city.

select dept,city,count(*) as empcount from employees group by city,dept order by dept,city ;

-- Q23. Find average salary grouped by department AND gender.

select dept,gender,avg(salary) as avgsalary from employees group by dept,gender order by dept;

-- Q24. Count employees grouped by region AND role.

select region,role,count(*) as empcount from employees group by region ,role order by region,role;

-- Q25. Find max salary grouped by state AND department.

select state,dept,max(salary) as maxsalary from employees group by state,dept order by state,dept;

-- Q26. Find total salary grouped by city AND gender.

select city,gender,sum(salary) as totalsalary from employees group by city,gender order by city;

-- Q27. Count employees per department, city, and show groups with more than 2 employees (HAVING).

select dept,city,count(*) as empcount from employees group by dept,city order by dept,city;

-- Q28. Find average salary grouped by region AND gender, ordered by region and avg salary descending.

select region,gender,avg(salary) as avgsalary from employees
group by region,gender order by region,avgsalary desc;

-- Q29. Count employees grouped by state AND role, show only combinations with more than 1 employee.

select state ,role,count(*) as empcount from employees
group by state,role having empcount > 1 order by state,role;

-- Q30. Count how many employees joined in each year.

```
select year(joindate) as joinyear, count(*) as empcount from employees group by joinyear order by joinyear;
```

-- Q31. Find average salary of employees grouped by the year they joined.

```
select year(joindate) as joinyear, avg(salary) as avgsalary from employees group by joinyear order by joinyear;
```

-- Q32. Count how many employees joined in each month (across all years).

```
select month(joindate) as joinmonth, count(*) as empcount from employees group by joinmonth order by joinmonth;
```

-- Q33. Find total salary grouped by quarter of the join date

```
select quarter(joindate) as Qtr, sum(salary) as totalsalary from employees group by Qtr order by Qtr;
```

-- Q34. Count employees who joined in Q1 (Jan–Mar) versus Q3 (Jul–Sep).

```
select quarter(joindate) as Qtr, count(*) as empcount from employees where quarter(joindate) in (1,3) group by Qtr order by Qtr;
```

-- Q35. Show the year and department with headcount, ordered by year and dept.

```
select year(joindate) as joinyear, dept, count(*) as empcount from employees group by joinyear, dept order by joinyear, dept;
```

-- Q36. Show years where more than 6 employees joined.

```
select year(joindate) as joinyear, count(*) as joiners from employees group by joinyear having joiners > 6 order by joinyear;
```

-- Q37. Find the average salary for employees joining in each quarter, show only quarters with avg salary above 70000.

```
select quarter(joindate) as Qtr, avg(salary) as avgsalary from employees group by Qtr having avgsalary > 70000 order by Qtr;
```

-- Q38. Find the number of employees and average salary per state.


```
select state, count(*) as empcount, avg(salary) as avgsalary from employees
group by state order by state;
```

-- Q39. Find the department with highest headcount in each state.

```
select state,dept,count(*) as empcount from employees group by state,dept order by
state,empcount desc;
```

-- Q40. Show cities with more than 2 departments represented.

```
select city,count(distinct dept) as deptcount from employees group by city having deptcount > 2;
```

-- Q41. Find the total salary bill per region AND state.

```
select region,state,sum(salary) from employees group by region,state order by region;
```

-- Q42. Show regions with more than 10 employees AND average salary above 67000.

```
select region,count(*) as empcount,avg(salary) as avgsalary from employees
group by region having empcount > 10 and avgsalary > 67000;
```

-- Q43. Count distinct roles available in each city.

```
select city,count(distinct role) uniquerole from employees group by city;
```

-- Q44. Count employees per department, but only consider female employees.

```
select dept,count(*) as femalecount from employees where gender = 'f' group by dept;
```

-- Q45. Find average salary per department for employees who joined after 2019-01-01.

```
select dept,avg(salary) as avgsalary from employees where joindate > 2019-01-01
group by dept order by avgsalary desc ;
```

-- Q46. Find total salary per city for employees in the IT department only.

```
select city,sum(salary) as totalsalary from employees where dept = 'it' group by city;
```

-- Q47. Count employees per role for those with salary above 70000.

```
select role,count(*) as empcount from employees where salary > 70000 group by role;
```

-- Q48. Show the max salary per department, only for employees in the South or West region.

```
select dept,max(salary) as maxsalary from employees where region in ('south','west') group by dept;
```

-- Q49. Count employees per year of joining for Texas (TX) state only.

```
select year(joindate) as joinyear, count(*) as empcount from employees where state = 'TX'  
group by joinyear order by joinyear;
```

-- Q50. Find departments where more than 3 female employees have salaries above 60000.

```
select dept, count(*) as female from employees where gender = 'f' and salary > 60000  
group by dept having female > 3;
```